

What Makes a 5-Star Restaurant?

A DS 4002 Case Study by Emily Friedman

Every day, millions of people rely on online reviews to decide where to eat. Platforms like Yelp have transformed the way customers evaluate restaurants, as they turn individual opinions into powerful signals that may determine a business's success or failure. A single star difference in rating can significantly impact customer traffic, revenue, and reputation.

Behind every rating is something more complex: **text**. Customers don't just leave scores. They write detailed reviews describing their experiences. These reviews contain valuable information about food quality, service, ambiance, and overall satisfaction. However, because reviews are unstructured, they can be difficult to systematically analyze and interpret.

This is where data science comes in.

You are a data scientist working for Yelp. Your goal is to uncover what drives restaurant ratings by analyzing customer reviews. Specifically, you will explore how the words and language used in reviews relate to a restaurant's rating. By identifying patterns in review text, you can help Yelp better understand customer preferences and provide more meaningful insights to users and businesses.

You will work with a dataset of Yelp reviews primarily from restaurants that serve dessert. You will process and analyze the text data to identify key words, phrases, and sentiment patterns that are associated with different rating levels. Ultimately, you will use your findings to answer an important question:

What factors, as expressed through customer reviews, most strongly predict whether a restaurant receives a high rating?

Your final deliverable will involve analyzing review text, identifying meaningful patterns, and communicating your findings clearly. You will take on the role of a data scientist, transforming raw text into actionable insights that explain what makes a restaurant successful.